THE GEORGE WASHINGTON UNIVERSITY

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Multimodal Financial Stock Price Forecasting

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Motivation & Problem Statement

- Markets move on numbers AND narratives — price history alone misses critical signals
- No clear consensus on which models + modalities actually drive better predictions
- Comparative study: single-modality baselines → full multimodal fusion
- Three modalities: price time series, financial news, quarterly earnings reports
- GAT-based models to capture cross-stock relationships
- Goal: 5-day price forecast across all combinations — find what works and why

Dataset: FinMultiTime (S&P 500 Subset)

Source: FinMultiTime — a large-scale multimodal financial dataset (Xu et al., 2025)

Coverage: 4,694 S&P 500 constituent stocks | 2009–2025

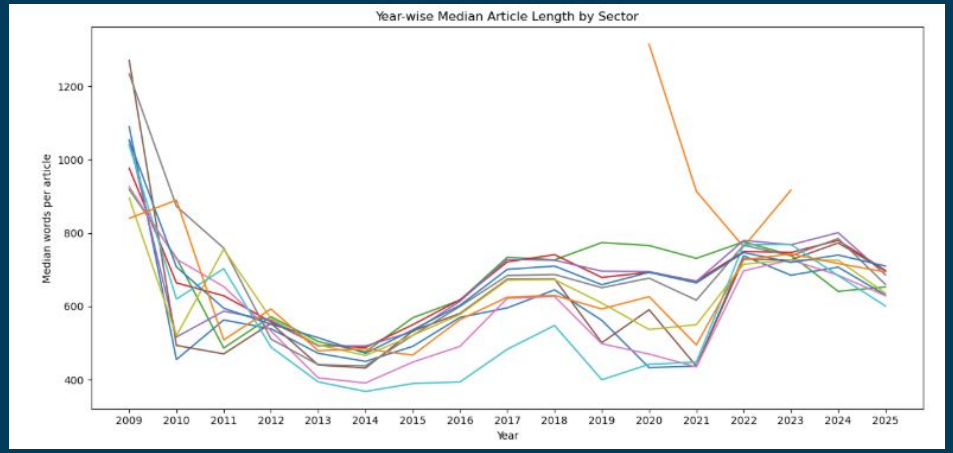
Three modalities used:

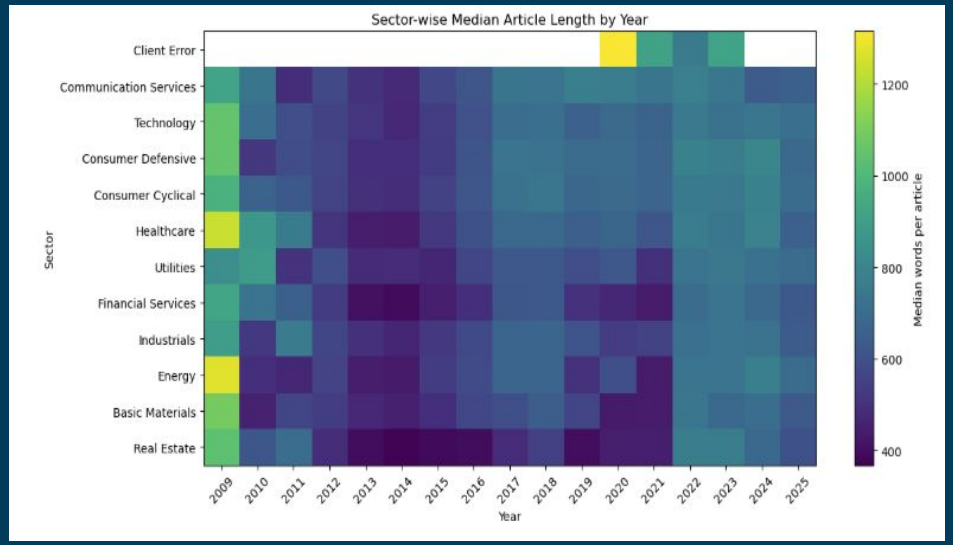
- Time Series — Daily OHLCV prices, normalized per-stock, 1.83 GB (CSV)
- News — Nasdaq articles with LSA summarization, 14.1 GB (JSONL)
- Financial Reports — SEC 10-K/10-Q filings covering balance sheet, cash flow, and equity statements, 84.04 GB (JSON/JSONL)

Exploratory Data Analysis

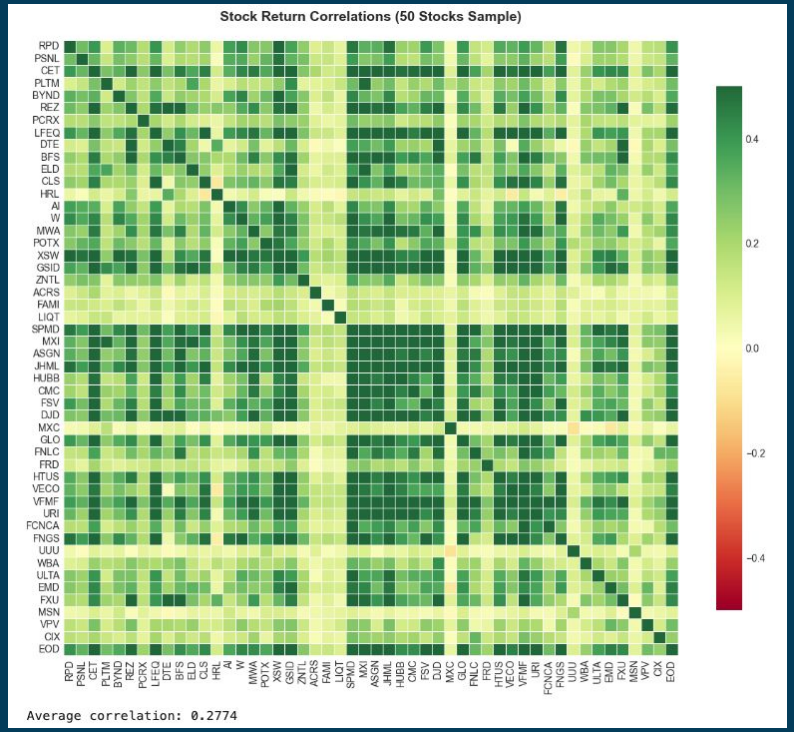
In our temporal EDA we tried to understand the data by looking at both news and time-series data. For our news data we tried to look at the following:-

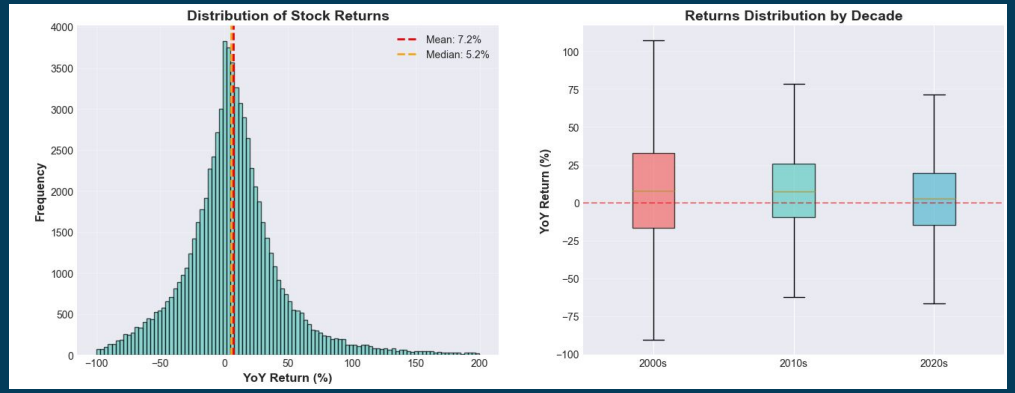
- Distribution of article length over time
- Sector wise median article length by year





Exploratory Data Analysis





- Time series data to understand the sector wise correlation and stock return

Methodology

Temporal Fusion Transformer (TFT)

Preprocessing & Model Architecture

1. **Preprocessing**

a. 5 base features (OHLCV) + 7 technical indicators (Bollinger Bands, RSI, MACD) = 12 features.
b. Standard scaling on training data.
c. NaN handling via forward fill , backward fill for rolling window edges.

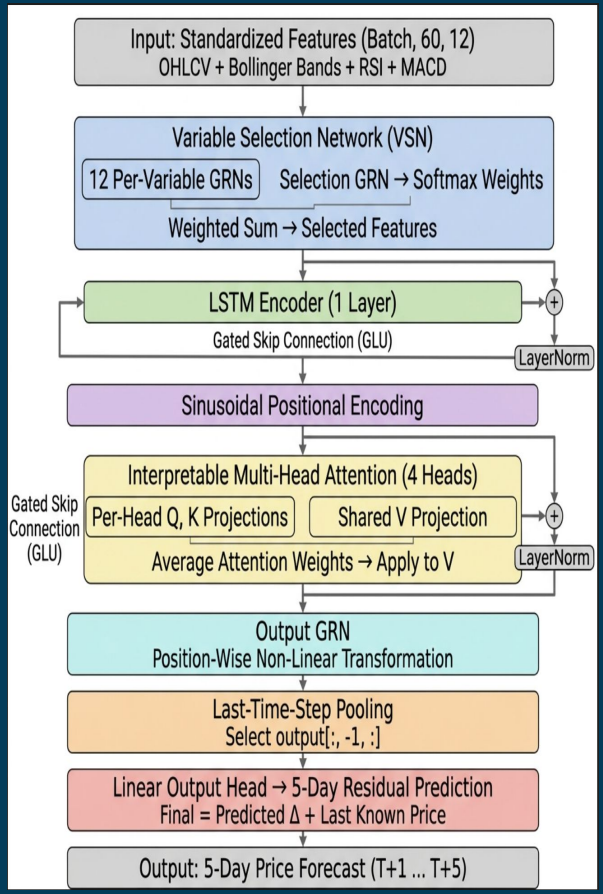
2. **Model**

a. **Variable Selection Network:** Automatically filters out noise to weigh the most crucial features at every time step.

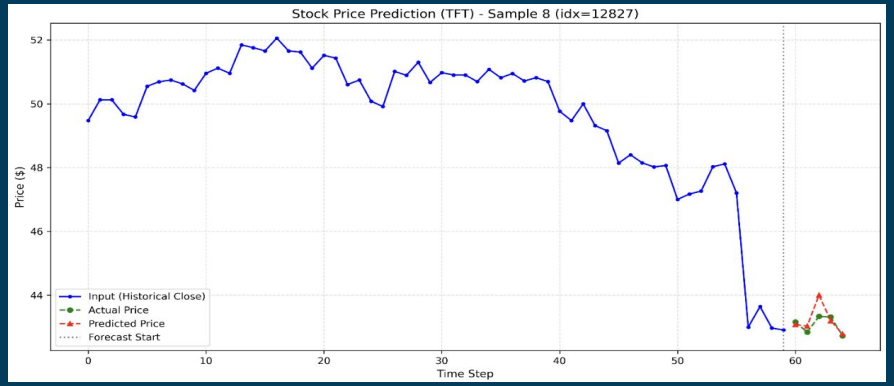
b. **LSTM + Attention Layers:** Combines short-term sequences with long-range pattern recognition.

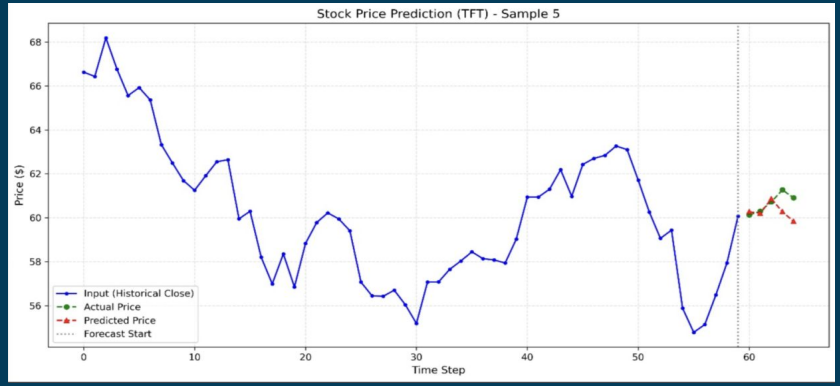
c. **Gated Residual Network:** Uses adaptive gates to dynamically control information flow and skip unnecessary processing layers.

d. **Multi-Step Forecasting:** Directly outputs a continuous 5 day future price trajectory in a single pass.



Temporal Fusion Transformer (TFT) - RESULTS





Metric	Value
MSE	0.0046
RMSE	0.0681
MAE	0.0357
MAPE	153.61%
SMAPE	10.29%

LSTM

Preprocessing & Model Architecture

1. Preprocessing

a. **Target**: Predict next-day stock return from OHLCV data; features use only past information to avoid data leakage.

b. **26 Engineered Features**: Returns, price patterns, volume trends, rolling statistics, technical indicators (RSI, ATR, MACD, Bollinger), and lagged returns.

c. **Split & Scaling**: Chronological 70/15/15; RobustScaler fit on train only.

d. **Windows**: 60-day input → 5-day forecast

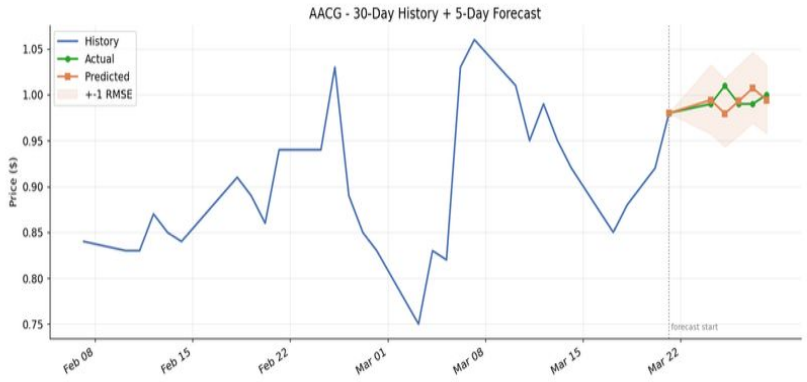
2. Model

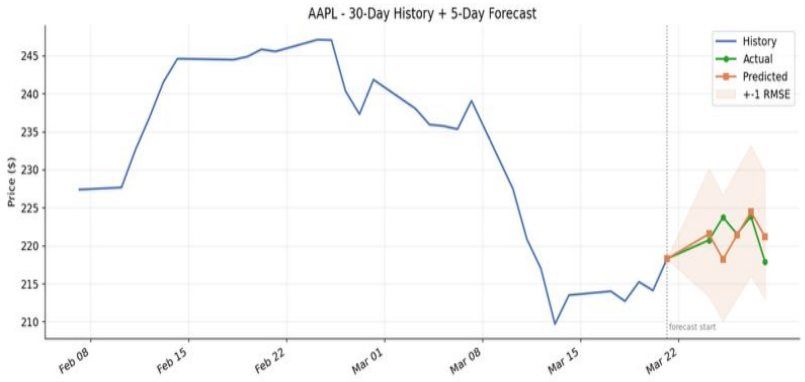
a. Two stacked LSTM layers (128 → 64 units) learn patterns at different time scales, with dropout to prevent overfitting (~130K parameters).

b. The model reads the full 60-day history and summarizes it into a single snapshot before making predictions.

c. Forecasts all 5 days at once in one step, rather than predicting one day and feeding it back, which prevents errors from piling up

LSTM Results





METRIC	VALUE
MAE	0.02902
RMSE	0.03750
MAPE	206.04%

TSMixer

Preprocessing & Model Architecture

1. Preprocessing

a. **Universe Selection**: Top 500 most liquid US stocks (2020–2025), ranked by median daily trading volume, with at least 1 year of history.

b. **15 Engineered Features**: Returns, price ranges, volume, volatility, technical indicators (RSI, ATR), plus calendar signals (day-of-week, month, month-end).

c. **Split & Scaling**: Chronological 70/15/15 per ticker; only the target is scaled with RobustScaler to keep evaluation metrics on original scale.

d. **Pooled Training**: All 500 tickers combined into one dataset using 60-day input → 5-day forecast windows

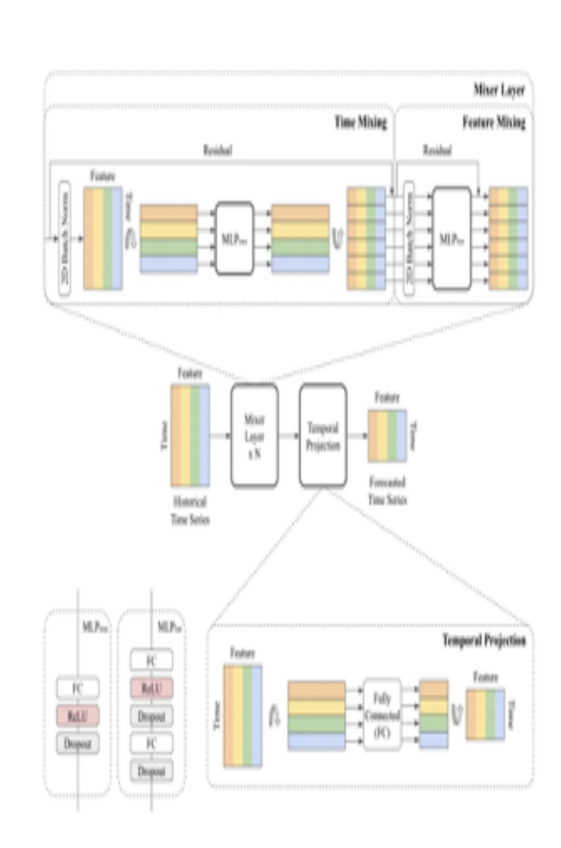
2. Model

a. **Architecture**: Four mixer blocks that alternately blend information across time and across features; a small ticker embedding adds stock-specific context

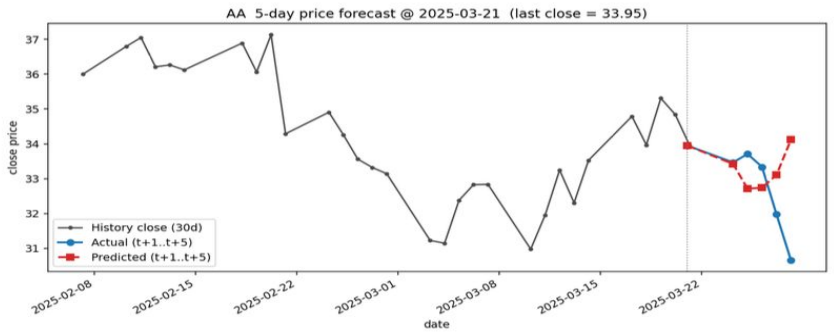
b. **RevIN Normalization**: Stabilizes training under shifting market conditions while keeping outputs on the original scale

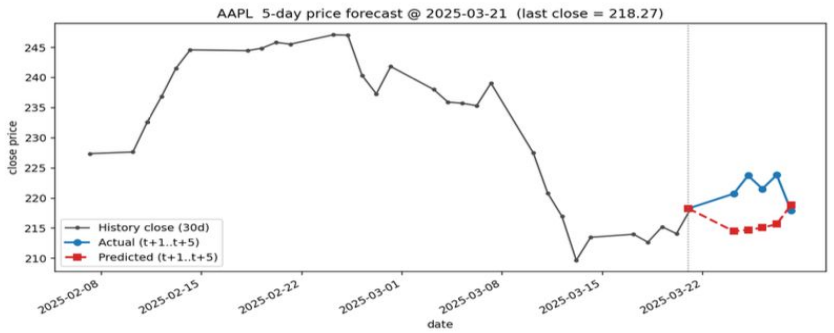
c. **Learnable Recency Weights**: Recent days start with higher importance, then adapt during training

d. **Direct Multi-Step Forecast**: Predicts all 5 days at once, avoiding step-by-step error buildup



TSMixer Results





METRIC	VALIDATION	TEST
MAE	0.0148	0.0147
RMSE	0.0238	0.0223
MAPE	26.00%	19.76%
Directional Accuracy	57.01%	51.29%

PatchTST

Preprocessing & Model Architecture

1. Preprocessing

a. **Data Cleaning**: OHLCV data cleaned, converted to numeric, and filtered for missing values before feature engineering

b. **Chronological Split**: 70/10/20 train/val/test per stock; val and test start 60 steps early to preserve lookback without leaking future data

c. **Global Scaling**: One StandardScaler fit across all training data normalizes inputs and target; predictions converted back to real returns at evaluation

d. **Training Setup**: AdamW with cosine schedule, Huber loss, gradient clipping, and early stopping on validation

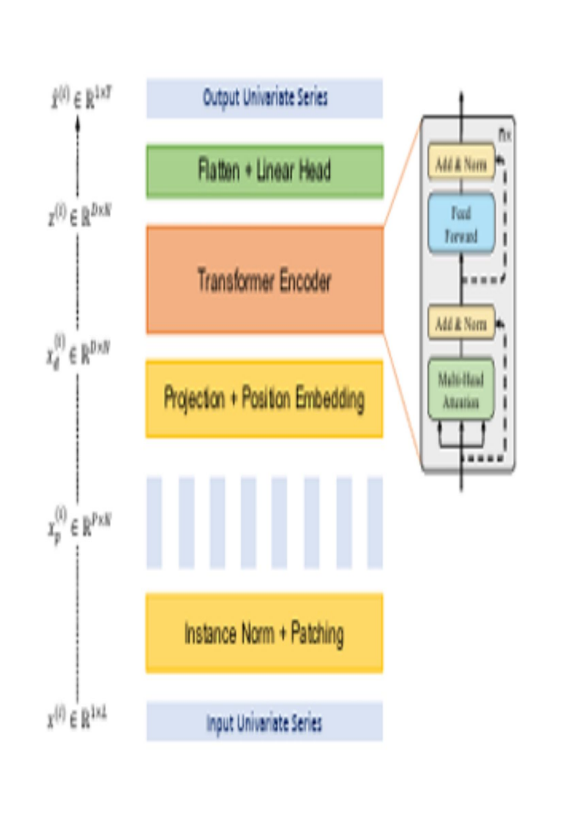
2. Model

a. **Channel-Independent Patching**: Each feature is processed separately and split into overlapping patches, letting the model focus on one signal at a time

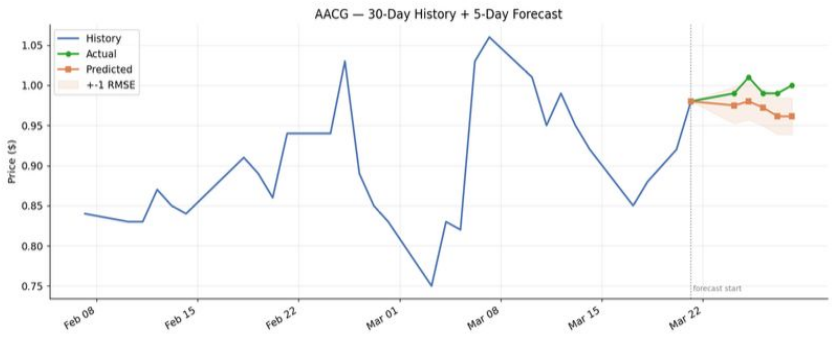
b. **Patch Encoding**: Patches projected into a 128-dim space and passed through 3 Transformer blocks with 8 attention heads (~410K parameters)

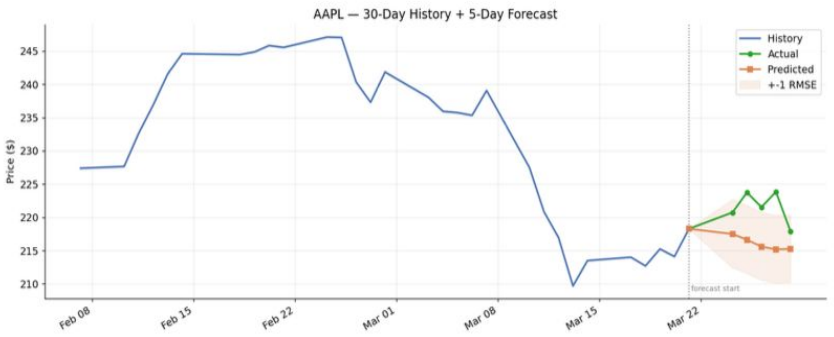
c. **RevIN Normalization**: Each channel is normalized before encoding and reverted after prediction, keeping outputs on the original scale

d. **Target-Only Head**: Only the log-return channel is mapped to the 5-day forecast through a linear head



PatchTST Results





Lightweight FinBERT & Multimodal FinBERT - TCN

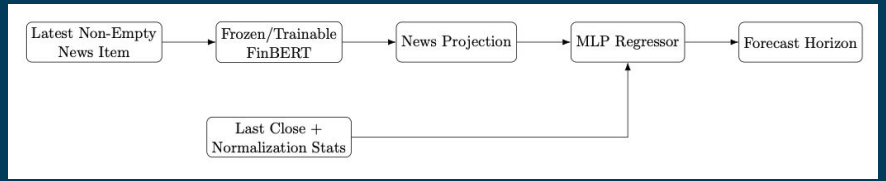
a. Text Processing - Financial news tokenized using FinBERT tokenizer with max length = 512 (padding + truncation), lowercase, punctuation removal

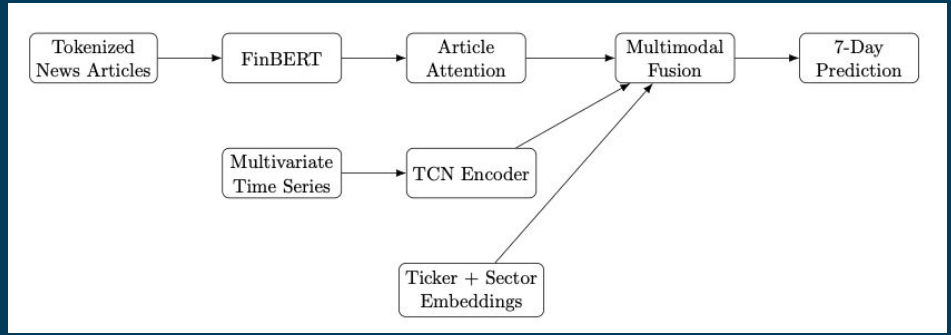
b. Missing Data Handling - Numerical values → mean imputation and no NaNs during training

c. Target Preparation - Targets normalized using input statistics, Maintains consistent scale between input/output

Experiment 2 tests whether a lightweight, efficient architecture can still extract useful predictive signal from financial news by using only the most recent news item and a shallow MLP regressor, rather than modeling the full article history.

Experiment 3 extends this into a richer multimodal setup, combining FinBERT-encoded news with a Temporal Convolutional Network for market data, plus learned ticker and sector embeddings, to test whether broader contextual fusion improves forecasting performance.





Result plots for FinBERT

Here are some plots for various experiments we did on FinBERT based models

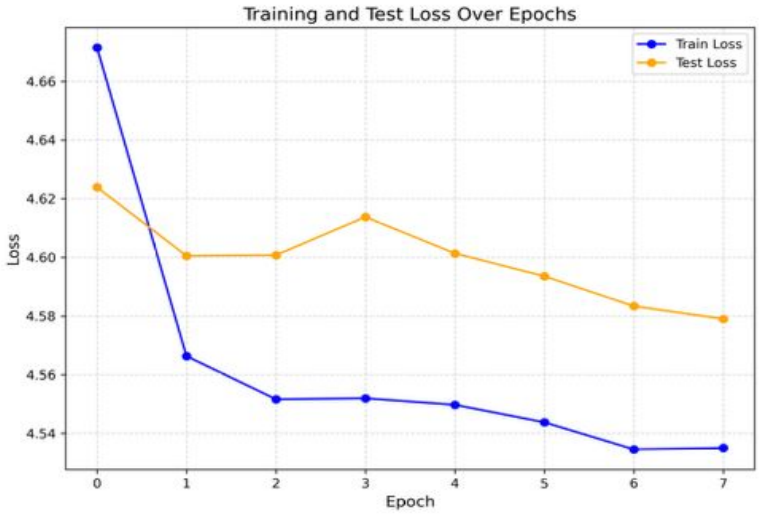
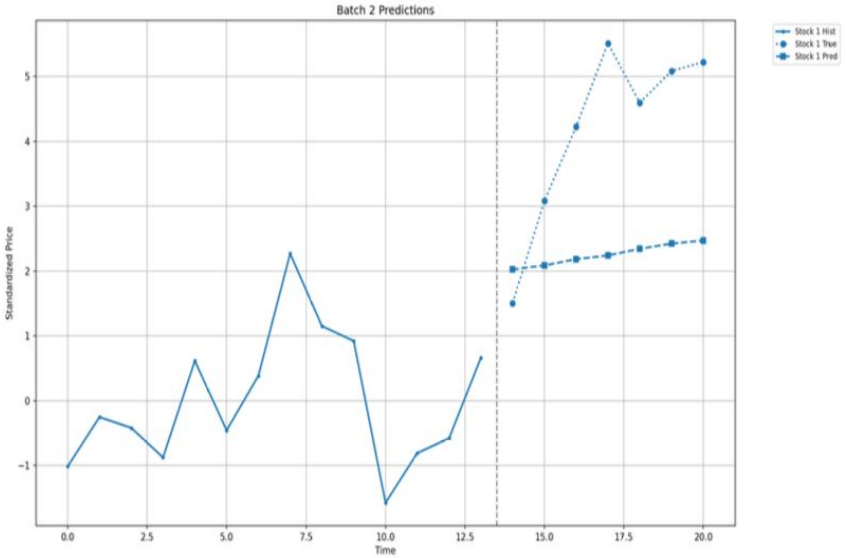


Figure 3: Training and test loss curves for Experiment 2.



Results for Lightweight FinBERT & Multimodal FinBERT–TCN

Metric	Exp 2 — Lightweight	Exp 3 — Multimodal TCN
Train MSE	1.75	4.64
Train RMSE	1.32	2.15
Train MAE	0.87	1.49
Test MSE	5.86	5.07
Test RMSE	2.42	2.25
Test MAE	1.63	1.53
Train–Test Gap (MSE)	4.11	0.43

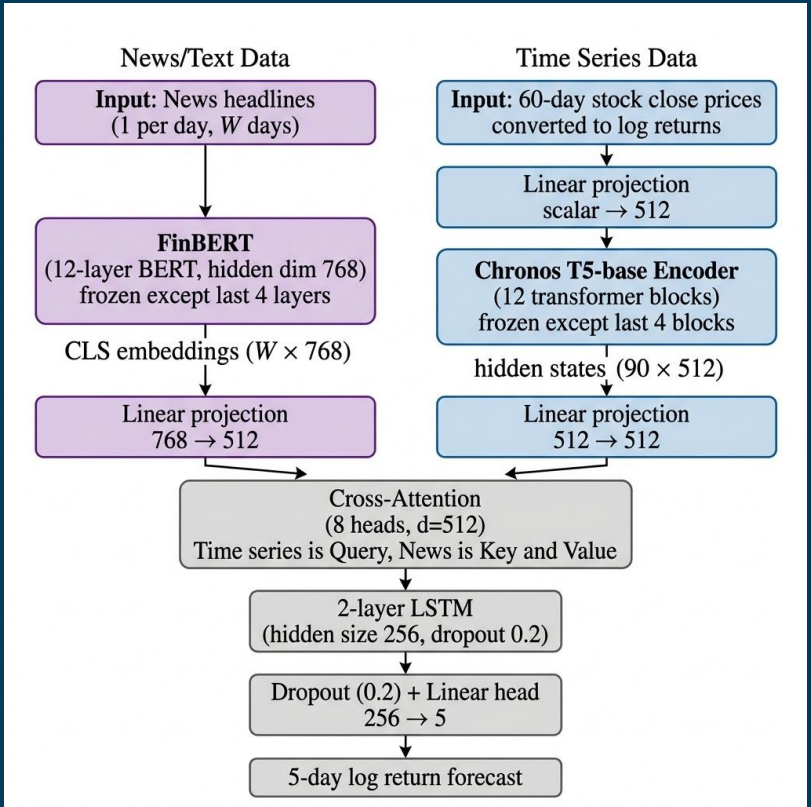
Multimodal Chronos T5 with FinBERT

Finbert : Preprocessing

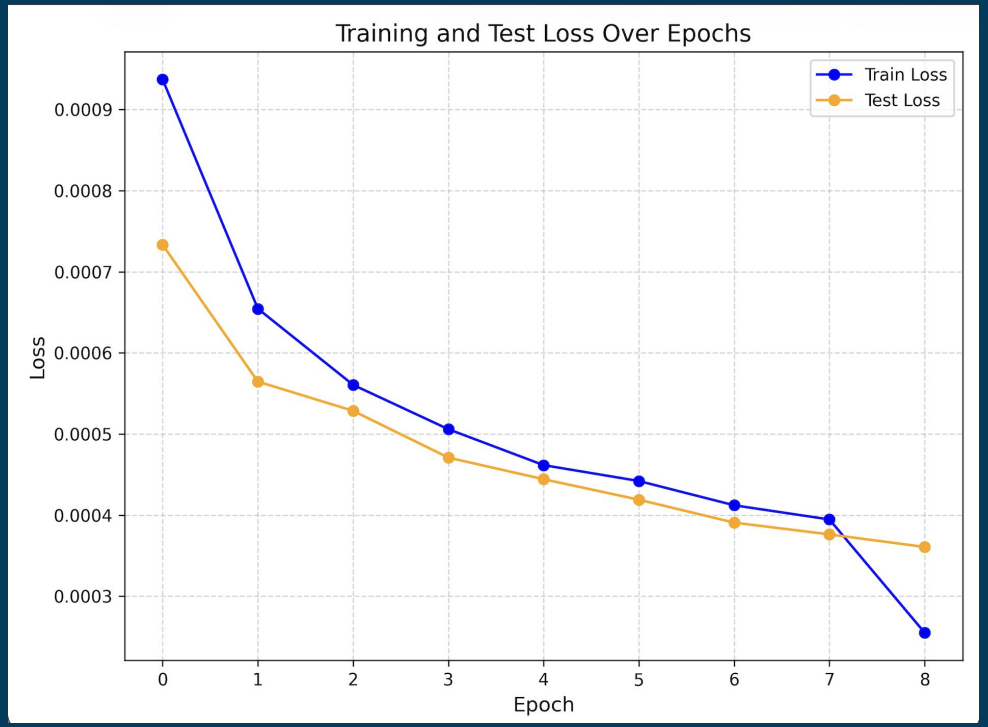
a. Text Processing - Financial news tokenized using FinBERT tokenizer with max length = 512 (padding + truncation), lowercase, punctuation removal

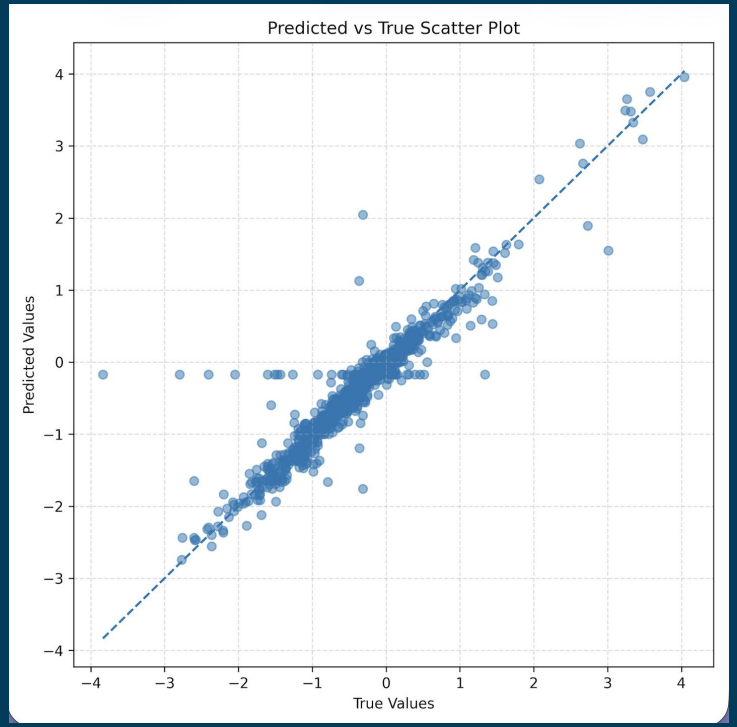
b. Missing Data Handling - Numerical values → mean imputation and no NaNs during training

c. Target Preparation - Targets normalized using input statistics, Maintains consistent scale between input/output



Results


Training and Test Loss Over Epochs
Train Loss
Test Loss
Loss
0.0009
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
0
1
2
3
4
5
6
7
8
Epoch


Predicted vs True Scatter Plot
Predicted Values
True Values
4
3
2
1
0
−1
−2
−3
−4

Chronos T5 - Finbert Multimodal

Metric	Chronos - FinBERT
Train MSE	1.75
Train RMSE	1.32
Train MAE	0.87
Test MSE	5.86
Test RMSE	2.42
Test MAE	1.63
Train–Test Gap (MSE)	4.11

TFT-Finbert Multimodal

Preprocessing & Model Architecture

1. Preprocessing

a. **News:** 1 article per day across 14-day window, tokenized via FinBERT tokenizer.
b. **Time series:** OHLCV + 7 technical indicators (Bollinger bands, RSI, MACD)
c. Targets standardized per-sample

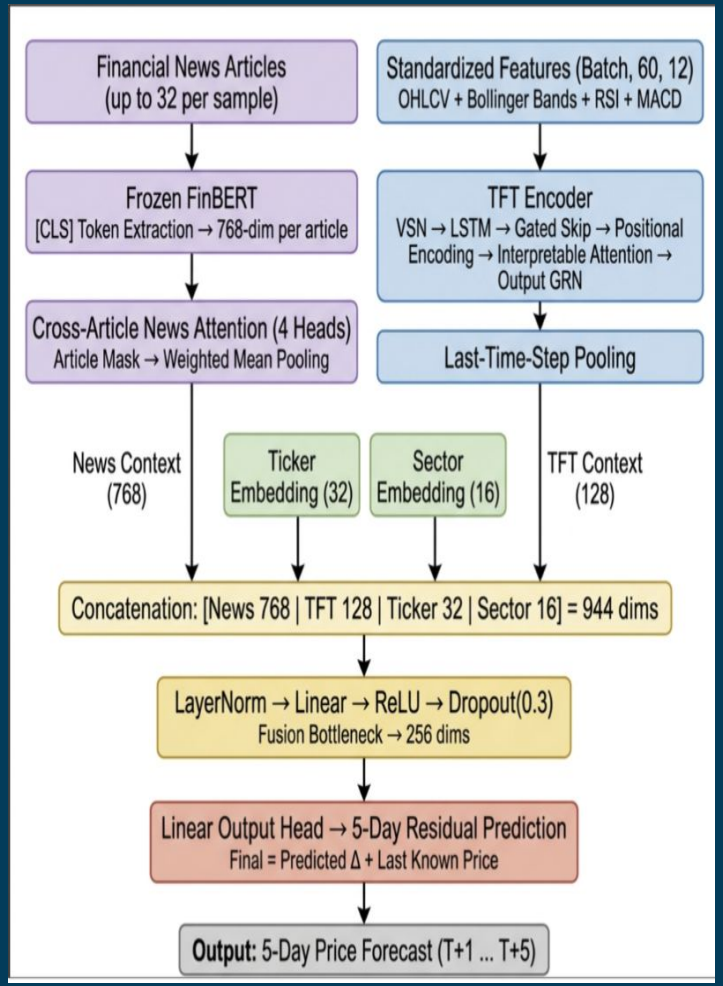
2. Model

a. **News Attention Pooling:** Uses FinBERT and Multi-Head Attention to dynamically weigh and merge multiple daily articles into a single sentiment vector.

b. **Temporal Context Extraction:** Slices out the final state from the temporal encoder to capture only the most up-to-date market conditions.

c. **Concatenated Fusion Bottleneck:** Merges the news vector, time-series state, and stock IDs into one unified tensor for joint processing.

d. **Raw Price Forecasting:** Translates the completely fused data into a final output that directly predicts the raw stock prices for the next 5 continuous days.



TFT-FINBERT Multimodal Results

Metric	Value
MSE	0.659
RMSE	0.812
MAE	0.521
sMAPE	59.23%
MAPE	176.85%

Results and Analysis

Future Work

- **Adaptive Fusion**: Teach the model *when* to trust each data source — trust sentiment on earnings days, ignore it on quiet news days.
- **Dynamic Relational Graphs**: Move beyond fixed sector links to evolving connections built from rolling correlations, supply chains, or news co-mentions
- **Full Multimodal Integration**: Bring chart images back into the pipeline to test if visual patterns add value beyond numerical price data
- **Regime-Aware Modeling**: Adapt the model to different market conditions — calm periods, earnings seasons, and crises behave very differently
- **Action-Based Evaluation**: Measure real-world usefulness through ranking quality and portfolio backtests, not just forecast error

Velocity/ Work Done

Research Paper

Multimodal Techniques for Financial Time-Series Forecasting

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Amir Jafari
George Washington University

April 2026

Abstract

Forecasting short-horizon stock movements remains difficult because market prices react to both persistent temporal structure and irregular external events. This paper condenses a larger benchmarking study on S&P 500 equities that compares unimodal, multimodal, and graph-based neural forecasting systems using aligned market prices, news sentiment, and structured fundamentals from SEC filings. The study builds a reduced U.S.-focused version of a large multimodal financial dataset and evaluates recurrent, convolutional, transformer, foundation-model, and graph-attention architectures under a common forecasting task. The benchmark spans classical sequence models, modern token-mixing architectures, pretrained time-series foundation models, and several fusion strategies for integrating textual context. The main finding is that bimodal models combining price trajectories with news sentiment yield the most reliable gains over price-only baselines, while graph models provide useful complementary structure through sector relationships. The results suggest that added complexity is worthwhile when it encodes genuinely complementary information rather than merely increasing parameter count.

1 Introduction

Financial forecasting has long depended on historical price data, yet market behavior is also shaped by news, investor sentiment, and changes in firm fundamentals. Recent multimodal datasets make it possible to study how these heterogeneous signals interact, but they also raise a practical question: which modalities and model families deliver meaningful predictive improvements for short-horizon forecasting?

This paper studies that question for S&P 500 stocks. Given a lookback window of historical prices and optional aligned textual and tabular inputs, the task is to predict the next 5–7 trading-day close prices. The paper presents a broad experimental comparison in which simple temporal baselines, multimodal fusion architectures, and graph neural networks are evaluated within one unified pipeline. The analysis emphasizes the experimental design, the role of cross-modal fusion, and the empirical patterns that matter most for practitioners.

The study is motivated by a central applied concern in financial machine learning: richer data do not automatically translate into better forecasts. Each additional modality introduces its own cadence, noise characteristics, missingness pattern, and alignment burden. As a result, a useful benchmark must evaluate not only whether multimodal learning helps, but also which fusion strategies are robust enough to justify their complexity.

A second motivation is methodological. In many financial applications, predictive pipelines are judged on narrow benchmark gains without sufficiently analyzing why a given model works, under which information conditions it fails, and whether its gains are likely to be operationally meaningful. By broadening the comparison to recurrent networks, patch-based transformers, MLP-style mixers, pretrained time-series foundation models, fusion networks, and graph architectures, the study moves beyond a single-model narrative and instead treats multimodal forecasting as a design space with explicit trade-offs.

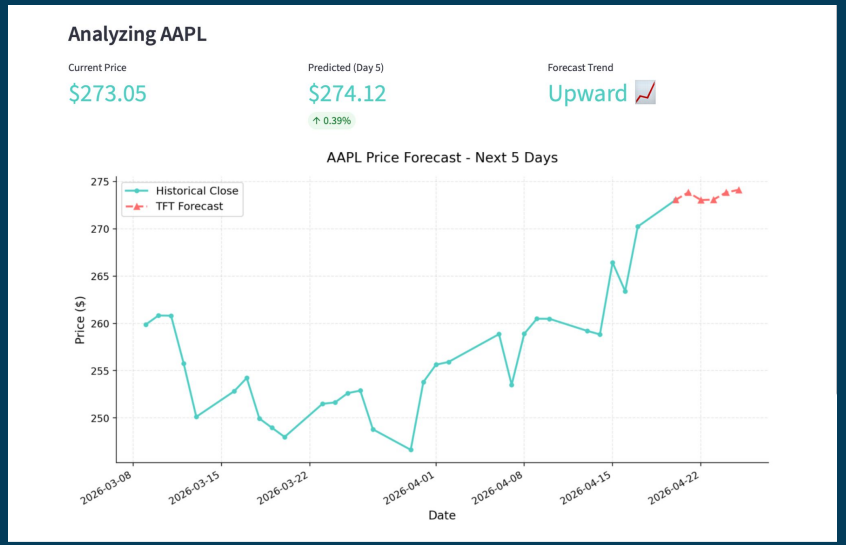
Finally, the work is relevant because the target problem is deliberately short horizon. Over a five- to seven-day window, market dynamics are heavily affected by event timing, sentiment shocks, and sector-level co-movement. This setting is therefore well suited to evaluating whether textual and relational context can complement price history in a statistically useful and operationally plausible way.

2 Data and Task Setup

The benchmark is built from the S&P 500 portion of a larger multimodal financial dataset. The full source includes price time series, financial news, corporate filings, and chart images. This paper restricts attention to the U.S. market and excludes image inputs in order to focus on structured and textual signals that can be aligned directly to the forecasting target.

All modalities are mapped to a daily trading calendar at the ticker level. Price-derived variables provide the dense temporal backbone. News is aggregated by publication date and merged into the daily frame, while filing-based fundamental variables are backward-aligned and forward-filled until a newer filing becomes available. After align-

DEMO APP



Forecast Data

	Date	Predicted Price
0	2026-04-21	$273.84
1	2026-04-22	$273.04
2	2026-04-23	$273.08
3	2026-04-24	$273.84
4	2026-04-25	$274.12

Thank You!